CHAPTER 4

DESIGNING

4.1 METHODS OF DESIGN OF REINFORCED CONCRETE:

The aim of design is to decide the size of the member and amount of reinforcement required, so that the structure will perform satisfactorily during its life period with minimum cost. The following three methods have been developed for the design of reinforced concrete structures.

a) Working stress method
b) Ultimate load method
c) Limit state method

4.1.1 Working stress method:

Working stress method is based on elastic theory assuming reinforced concrete as elastic material. The stress strain curve of concrete is assumed as linear from zero at the neutral axis to a maximum value at the extreme fiber.

This method adopts permissible stresses which are obtained by dividing ultimate stress by a factor known as factor of safety. For concrete a factor of safety of 3 is used and for steel it is 1.78. This factor of safety account for any uncertainties in estimation of working loads and variation in material properties.

In working stress method, the structural member is designed for working loads such that the stresses developed are within the allowable stress. Hence, the failure criterion is the stress. This method is simple and reasonably reliable.

The drawbacks of this method are

1) Stress strain curve for concrete is assumed as linear, which is not true.
2) Factor of safety does not predict the true margin of safety.
3) The failure criteria assumed is stress but strain criteria is the reliable.
4) The effect of the creep and shrinkage of the concrete is ignored.
5) This method gives uneconomical section.

This method has been deleted in IS: 456-2000, but the concept of this method is retained for checking the serviceability state of deflections and cracking. Hence, the knowledge of this method is essential and IS: 456-2000 gives it in the appendix.

4.1.2 Ultimate load method:

In ultimate load method, structural elements are designed for ultimate loads which are obtained by multiplying the working loads with a factor known as load factor. Hence, the designer can able to predict the excess load the structure can carry beyond the working load

without collapse. Hence, this method gives the true margin of safety. This method considers the actual stress strain curve of concrete and the failure criteria is assumed as ultimate strain.

This method gives very economical sections. However it leads to excessive deformation and cracking. Thus, this method is failed to satisfy the serviceability and durability requirements. To overcome these drawbacks, the limit state method has been developed to take care of both strength and serviceability requirements.

4.1.3 Limit state method:

In the limit state method, the structural elements are designed for ultimate load and checked for serviceability (deflection, cracking etc) at working loads so that the structure is fit for use throughout its life period. The details of this method are given in article 2.1.

4.2 PHILOSOPHY OF LIMIT STATE DESIGN:

A structure may become unfit for use not only when it collapse but when it violates the serviceability requirements such as deflection, cracking etc. the philosophy of limit state method of design is to see that the structure remains fit for use throughout its life period by assuring safety against strength and serviceability requirements i.e. the structure will not reach the limit state in its life time. The acceptable limit for safety against strength and serviceability requirements before failure occurs is called limit state. All the relevant limit state has to be considered in the design. The load and strength of the materials are to be estimated by probabilistic approach (characteristic values). The design loads and strengths are derived from the characteristic values through the use of partial safety factors.

4.2.1 Limit states:

The various limit states to be considered in the design are

1) Limit state of collapse
2) Limit state of serviceability

Limit state of collapse:

It is the limit state at which the structure is likely to collapse. The structure may collapse due to rupture of one or more critical sections or loss of overall stability due to buckling or overturning. This limit state may correspond to

a) Flexure
b) Compression
c) Shear
d) torsion

Limit state of serviceability:

Limit state serviceability relate to the performance of the structure at working loads. It is the limit state at which the structure undergone excessive deflection, which adversely affect the finishes causing discomfort to the users and excessive cracking which effects the efficiency or appearance of the structure.

This limit state may correspond to

a) deflection
b) cracking
c) other limit states (vibrations, fire resistance, durability)

4.2.2 Characteristic strength of materials:

The characteristic strength of material is that value of the strength of the material below which not more than 5% of the test results are expected to fall.

For concrete, characteristic strength is compressive strength of 150 mm cubes of 28 days in N/mm2, below which not more than 5% of the test results are expected to fall. Concrete grade are specified based on this strength.

For steel, the minimum yield strength or 0.2% proof strength is taken as the characteristic strength of steel.

4.2.3 Characteristic loads:

The maximum working load that the structure has to withstand and for which it has to be designed is called characteristic load. The characteristic loads are calculated based on statistical analysis.

As per code, the characteristic load is defined as that value of load which has 95% probability of not being exceeded during the life of the structure.

As statistical data of loads is not available, dead loads given in IS:875 (Part-1), imposed loads given in IS:875 (Part-2),wind loads given in IS:875 (Part-3), snow loads given in IS:875 (Part-4) and seismic forces given in IS:1893 shall be assumed as the characteristic loads as per IS:456-2000.

4.2.4 Design value and partial safety factors:

Materials: the design strength of materials is obtained by dividing the characteristic strength by a factor known as partial safety factor. The partial safety factor takes in to account variation of material strength, local weakness etc.

The design strength of the materials, is given by

$$Design\ strength = \frac{Characteristic\ strength}{partial\ safty\ factor}$$

Material	Limit state of collapse	Limit state of serviceability
Steel	1.15	1.0
Concrete	1.5	1.0

Table 4.1: Partial safety factor for material strength

Loads: design loads or factored loads, for which the structure is to be designed, are obtained by multiplying the characteristic load with a factor known as partial safety factor. The partial safety factor takes in to account inaccurate assessment of load, unexpected stress redistribution and variation in dimensional accuracy.

Load combination	Limit state of collapse			Limit state of serviceability		
	DL	LL	WL	DL	LL	WL
DL+LL	1.5	1.5	-	1.0	1.0	
DL+WL	1.5 or 0.9	-	1.5	1.0	-	1.0
DL+LL+WL	1.2	1.2	1.2	1.0	0.8	0.8

Table 4.2: Partial safety factor for loads

4.3 CODE OF PRACTICE FOR R.C.C. DESIGN:

All reinforced concrete structural design in our country should conform to the revised Indian standard code IS: 456-2000, code of practice for plain and reinforced concrete (Forth revision).

The Bureau of Indian Standards have released over the years several hand books to facilitate reinforced concrete structural designers to design routine structural elements quickly by referring to the various tables and graphs presented in the hand books.

The following hand books will serve as useful design aids for structural concrete designers.

a) SP 16 – 1980 – Design aids for reinforced concrete to IS:456
b) SP 24 – 1983 – Explanatory hand book on IS:456
c) SP 34 – 1987 – Hand book on concrete reinforcement and detailing.
d) SP 23 – 1982 – Hand book on concrete Mixes.